TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

STUDENT NAME:			
STUDENT ID NUMBER:		TION NUMBER IN THIS COURSE:	

HW1 – FALL 2026

COURSE NAME:	*Big Data Analytics*
COURSE NUMBER:	SOC4170
EXAMINATION DATE:	TIME:
EXAMINATION DURATION:	90 Minutes
ADDITIONAL MATERIALS ALLOWED TO USE:	**Calculators are allowed**
SPECIAL INSTRUCTIONS:	*- Answers will only be evaluated if they are readable.* *- Show all your work. For some questions, you may get partial credit even if the end result is wrong due to a calculation mistake. If you make assumptions, state your assumptions clearly and precisely.*

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

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1.[37 points] Frequent Pattern Mining & Word Count

We have 4 transactions as follows;

T0 = {A, B, C, D}

T1 = {A, C, E, D}

T2 = {A, E, D}

T3 = {A, B, C, E, D}

The minimum support is 70% and the minimum confidence is 80%.

(a) [4 Points] Write frequent 1-itemsets.

Frequent 1-itemsets: { },

(b) [8 Points] Write 2-itemsets of self-join result. Also write candidate 2-itemsets.

2-itemsets of self-join result: { },

candidate 2-itemsets: { },

(c) [4 Points] Write frequent 2-itemsets.

Frequent 2-itemsets: { },

(d) [9 Points] Write 3-itemsets of self-join result. Also write candidate 3-itemsets and frequent 3-itemsets.

3-itemsets of self-join result: { },

candidate 3-itemsets: { },

frequent 3-itemsets: { },

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(e) [12 Points] Write all the rules with support and confidence.
Rule1: { } -> { }, xx%, yy%
Rule2: { } -> { }, xx%, yy%
…
where xx% is support, and yy% is confidence.

2.[25 points] Inverted Index & Page Rank

We have 4 documents as follows;

D0: BTS BLACK PINK MUSIC AWARD

D1: BLACK PINK COLOR

D2: SONG BLACK PINK

D3: MUSIC BTS

D0 has hyperlinks to D2

D1 has a hyperlink to D0, D3

D2 has a hyperlink to D0, D1

D3 has a hyperlink to D2

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(a) [4 Points] Let 4 Mappers M0 - M3 take documents D0 – D3 respectively, write the output (list of <key, value> pairs) of each mapper to come up with Inverted Index.

Output of M0 = < >,

Output of M1 = < >,

Output of M2 = < >,

Output of M3 = < >,

(b) [4 Points] We have 2 Reducers R0 and R1. R0 takes <key, (value1, value2, …)> pairs if the first character of the key is from 'A' to 'J'. R1 takes the rest. Write the output of each reducer to come up with Inverted Index.

Output of R0 = < >,

Output of R1 = < >,

(c) [4 Points] The equation to get the page rank of a document is as follows;

$$r_i = d\left[\sum_{p_j \in \mathcal{I}(p_i)} \frac{r_j}{c_j}\right] + (1-d)\frac{1}{|D|}$$

Where d = 0.8. And d is multiplied only by reducers.
Each document's initial page rank is 10
Let 4 Mappers M0 - M3 take documents D0 – D3 respectively, write the output (list of <key, value> pairs) of each mapper to come up with the page rank of each document. Round off the number to the nearest tenth. For example, 6.65 is 6.7 and 3.33 is 3.3.

Output of M0 = < >,
Output of M1 = < >,
Output of M2 = < >,
Output of M3 = < >,

(d) [4 Points] We have 4 Reducers R0 – R3. Write the output of each reducer to come up with page rank of each document. For simplicity, **ignore** the second part of the equation.
Round off the number to the nearest tenth. For example, 6.65 is 6.7 and 3.33 is 3.3.

Output of R0 = < >
Output of R1 = < >
Output of R2 = < >
Output of R3 = < >

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(e) [8 Points] Assume that we need one more cycle of Map/Reduce to come up with the page rank of each document with the same number of mappers and reducers. Write the output of mappers and reducers. Again, for simplicity, **ignore** the second part of the equation.
Round off the number to the nearest tenth. For example, 6.65 is 6.7 and 3.33 is 3.3.

Output of M0 = < >,
Output of M1 = < >,
Output of M2 = < >,
Output of M3 = < >,

Output of R0 = < >
Output of R1 = < >
Output of R2 = < >
Output of R3 = < >

(f) [1 Points] Now, we have the page rank of each document. Write the sequence of documents when a user typed BLACK PINK in the search window and clicked "search" button.

3.[7 points] The Recommendation System

"Inha Cinema" is a company providing movie services. Currently "Inha Cinema" has 6 customers from C1 to C6.

C1 has watched M1, M4, M6. C2 has watched M2, M3, M5, M6. C3 has watched M2, M4, M5, M6. C4 has watched M1, M4, M5, M6. C5 has watched M4, M5, M6. C6 has watched M3, M4, M5, M6. "Inha Cinema" is preparing a movie recommendation list to C5 by finding 3 most similar customers to C5 and by finding movies which 3 most similar customers have watched but C5 has not watched. Jaccard coefficient is used to measure the similarities between customers.

$$\text{Jaccard}(C_i, C_j) = \frac{|C_i \cap C_j|}{|C_i \cup C_j|}$$

(a) [5 Points] Calculate Jaccard coefficients between C5 and C1-C6.

Jaccard(C5, C1): **Jaccard(C5, C2):** **Jaccard(C5, C3):**

Jaccard(C5, C4): **Jaccard(C5, C6):**

(b) [2 Points] List movies of the movie recommendation list to C5

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4.[15 points] Decision Tree

The table below represents a training set, D, of class-labeled tuples. The class label "buys_computer" has two distinct values (yes, no). The attribute list contains "age", "income", "student", "credit_rating".

RID	*age*	*income*	*student*	*credit_rating*	*Class: buys_computer*
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no

And the table below represents $\log_2 n$ from n=1 to n=18

N	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**
$\log_2 n$	**0**	**1**	**1.6**	**2**	**2.3**	**2.6**	**2.8**	**3**	**3.2**	**3.3**	**3.5**	**3.6**	**3.7**	**3.8**	**3.9**	**4**	**4.1**	**4.2**

$$Info(D) = -\sum_{i=1}^{m} p_i \log_2(p_i) \qquad Info_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Info(D_j) \qquad Gain(A) = Info(D) - Info_A(D)$$

(a) [2 Points] What is Info(D)? Show all your work.

(b) [8 Points] What is $Info_{age}(D)$ and $Info_{income}(D)$? Show all your work.

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(c) [4 Points] What is $Gain_{age}(D)$ and $Gain_{income}(D)$? Show all your work.

(d) [1 Points] Which attribute can be the splitting criterion between age and income?

5.[16 points] Confusion Matrix

(a) [8 Points] We have a model M1 which detects fraud credit card transactions. Total number of test data transactions are 1000 and the test data contains 5 actual fraudulent transactions. M1 detected 10 fraudulent transactions, 3 of which are real fraudulent transactions. Complete the confusion matrix and compute accuracy, precision, and recall.

Predicted class Actual class	Fraud = Yes	Fraud = No	Total
Fraud = Yes			
Fraud = No			
Total			

Accuracy: % (2 points)

Precision: % (2 points)

Recall: % (2 points)

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(b) [8 Points] We have a model M2 which decides loan approvals. Total number of loan reviews are 1000 and the test data contains 502 loan approvals. M2 detected 500 loan approvals, 499 of which are real loan approvals. Complete the confusion matrix and compute accuracy, precision, and recall.

Predicted class Actual class	Loan = Yes	Loan = No	Total
Loan = Yes			
Loan = No			
Total			

Accuracy: % (2 points)

Precision: % (2 points)

Recall: % (2 points)